

A photograph of several Emperor penguins swimming underwater in clear, blue water. The penguins are seen from below, showing their dark backs and white bellies. They are moving in various directions, with some leaving visible trails of bubbles behind them. The lighting is bright, creating a high-contrast scene.

Vapor trails

Emperor penguins dive into the icy waters off Antarctica in search of food. Above water, they hold a layer of insulating air trapped under their feathers, but this streams off as a trail of bubbles behind the swimming birds. Emperors have a top speed of more than 15 mph (25 kph). They swim by beating their stiff wings and use their webbed feet to steer. Unlike flying birds, they have heavy bones, which makes it easier for them to plunge into the depths. Emperors feed on fish and krill, and they can dive to a depth of more than 1,650 ft (500 m), holding their breath for up to 20 minutes.